

ADECCO HR ANALYTICS – CASE STUDY REPORT

1. Executive Summary

Human Resources (HR) plays a critical role in organizational success by ensuring employee satisfaction, engagement, productivity, and retention. Employee attrition is one of the most significant challenges faced by organizations because it increases recruitment costs, disrupts operations, reduces productivity, and leads to the loss of valuable talent.

The Adecco HR Analytics project aims to analyze employee data to identify workforce trends, understand factors influencing employee attrition, and provide actionable recommendations for improving employee retention and organizational performance.

Using Excel-based analytical techniques such as PivotTables, charts, correlation analysis, scatter plots, and statistical summaries, this project evaluates employee demographics, compensation, job satisfaction, work-life balance, training effectiveness, promotion opportunities, and other workforce-related factors.

The insights generated from this analysis help management make data-driven decisions that improve employee engagement, reduce turnover, and strengthen workforce stability.

2. Problem Statement

Employee attrition is a major concern for organizations because replacing experienced employees requires significant time, effort, and financial investment.

The primary business challenge is:

To identify the key factors influencing employee attrition and workforce performance while developing strategies that improve employee satisfaction, retention, productivity, and long-term organizational growth.

The project focuses on understanding employee behavior and workplace conditions that contribute to turnover and performance outcomes.

3. Business Objectives

The main objectives of this project are:

Employee Retention

Identify factors contributing to employee attrition and reduce turnover.

Workforce Engagement

Analyze employee satisfaction and involvement levels.

Performance Improvement

Understand factors affecting employee performance ratings.

Career Development

Evaluate the impact of promotions, training, and tenure on employee satisfaction and retention.

Compensation Analysis

Assess the relationship between salary, stock options, and employee turnover.

Strategic Workforce Planning

Provide actionable recommendations to improve workforce management and organizational effectiveness.

4. Dataset Overview

The dataset contains employee information from Adecco India across various departments and job roles.

Key Categories of Data

Employee Demographics

- Age
- Gender
- Marital Status
- Education
- Education Field

Employment Information

- Department
- Job Role
- Business Travel
- Job Level
- Years At Company
- Years Since Last Promotion
- Years In Current Role

Compensation & Benefits

- Monthly Income
- Stock Option Level
- Percent Salary Hike

Employee Engagement

- Job Satisfaction
- Environment Satisfaction
- Relationship Satisfaction
- Work-Life Balance
- Job Involvement

Performance Metrics

- Performance Rating
- Training Times Last Year

Attrition Indicator

- Attrition (Yes/No)
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5. Methodology

The project follows a structured HR Analytics framework.

Step 1: Data Preparation

- Data cleaning
- Validation of records
- Handling missing values
- Formatting numerical variables

Step 2: Exploratory Data Analysis

- Workforce demographics
- Attrition analysis
- Compensation analysis
- Satisfaction analysis

Step 3: Statistical Analysis

- Correlation Analysis
- Trend Analysis
- Attrition Rate Calculation
- Comparative Analysis

Step 4: Visualization

- Pivot Charts
- Bar Charts
- Pie Charts
- Scatter Plots
- Trendlines

Step 5: Business Interpretation

- Insight extraction
 - Workforce recommendations
 - HR strategy formulation
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6. Analytical Techniques Used

The following Excel techniques were used:

PivotTables

Used for:

- Attrition analysis
- Job role comparison
- Department-wise analysis
- Satisfaction analysis

COUNTIF

Used for:

- Employee counts
- Attrition counts
- Overtime employee calculations

AVERAGE

Used for:

- Average income
- Average age
- Average tenure
- Satisfaction scores

Correlation Analysis

Used to:

- Identify factors most strongly associated with attrition

Scatter Plot Analysis

Used to:

- Analyze relationships between employee variables

Trendline Analysis

Used to:

- Identify positive or negative trends

7. Workforce Profile Analysis

The workforce consists of employees from multiple departments, job levels, and functional roles.

The employee population exhibits diversity in:

- Age groups
- Educational backgrounds
- Marital status
- Job responsibilities
- Experience levels

Understanding workforce composition is important for designing effective HR policies.

8. Employee Attrition Analysis

Attrition analysis is one of the primary focus areas of this project.

The study examined:

- Employee age
- Salary levels
- Job involvement
- Work-life balance
- Promotion frequency
- Training participation
- Stock option benefits

The findings indicate that attrition is influenced by a combination of compensation, engagement, career growth opportunities, and workplace satisfaction.

9. Key Findings from the Analysis

Finding 1: Average Age of Employees Leaving the Company

The average age of employees who left the company is approximately **34 years**.

Interpretation

This indicates that attrition is concentrated among employees in their early and mid-career stages.

Business Impact

Organizations risk losing employees during their most productive career years.

Finding 2: Job Satisfaction Varies Across Job Roles

Job satisfaction levels differ across job roles.

Roles such as:

- Healthcare Representative
- Research Scientist
- Sales Executive

show relatively higher satisfaction compared to some managerial and support roles.

Business Impact

Role-specific engagement programs may be required.

Finding 3: Male Employees Show Slightly Higher Attrition

Analysis shows:

- Female Attrition Rate $\approx$ 15%
- Male Attrition Rate $\approx$ 17%

Interpretation

Although the difference is not extremely large, male employees exhibit slightly higher turnover.

Business Impact

Retention initiatives should consider workforce demographics.

Finding 4: Employees Leaving Earn Lower Monthly Income

Average monthly income among employees leaving the organization is lower than overall workforce averages.

Interpretation

Compensation may influence employee retention decisions.

Business Impact

Competitive salary structures may improve retention.

Finding 5: Distance from Home Influences Attrition

Attrition increases as distance from home increases.

Distance Category Attrition Rate

Near	14%
Medium	20%
Far	22%

Interpretation

Long commutes negatively affect employee retention.

Business Impact

Flexible work arrangements may reduce turnover.

Finding 6: Majority of Employees Have Performance Rating 3

Performance ratings are concentrated around Rating 3.

Interpretation

Most employees meet performance expectations.

Business Impact

Organizations should create pathways for employees to achieve higher performance levels.

Finding 7: Overtime Employees Represent Significant Workforce Segment

Approximately 28% of employees regularly work overtime.

Interpretation

Extended workloads may contribute to burnout.

Business Impact

Workload balancing should be monitored.

Finding 8: Average Employee Tenure

Average tenure is approximately 7 years.

Interpretation

The organization demonstrates moderate employee loyalty.

Business Impact

Retention programs can further improve long-term workforce stability.

10. Mid-Level Analysis Insights

Job Involvement vs Attrition

Attrition decreases as job involvement increases.

Job Involvement	Attrition Rate
1	34%
2	19%
3	14%
4	9%

Conclusion

Employees who feel engaged and involved are significantly less likely to leave.

Marital Status vs Attrition

Status	Attrition Rate
Single	26%
Married	12%
Divorced	10%

Conclusion

Single employees exhibit the highest turnover.

Training vs Attrition

Employees receiving more training generally show lower attrition rates.

Conclusion

Professional development contributes positively to employee retention.

Stock Options vs Attrition

Stock Option Level Attrition Rate

0	24%
1	9%
2	8%
3	18%

Conclusion

Employees receiving stock benefits are generally less likely to leave.

Business Travel vs Job Satisfaction

Employees who travel frequently show similar satisfaction levels compared to non-travelers.

Conclusion

Business travel does not appear to have a major negative effect on satisfaction.

11. Major HR Insights

Employee Engagement Matters

Higher job involvement significantly reduces attrition.

Career Growth Is Critical

Promotion opportunities influence employee satisfaction and performance.

Compensation Influences Retention

Salary and stock options affect employee loyalty.

Work-Life Balance Supports Productivity

Employees with healthier work-life balance demonstrate stable performance.

Training Improves Retention

Continuous learning opportunities strengthen workforce commitment.

12. Strategic Recommendations

Recommendation 1: Strengthen Employee Engagement Programs

Implement:

- Recognition programs
 - Employee participation initiatives
 - Team-building activities
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Recommendation 2: Improve Career Development Opportunities

Focus on:

- Promotion transparency
 - Internal mobility programs
 - Leadership development
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Recommendation 3: Enhance Compensation Strategy

Review:

- Salary competitiveness
 - Incentive structures
 - Stock option programs
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Recommendation 4: Promote Work-Life Balance

Introduce:

- Flexible work schedules
 - Hybrid work options
 - Wellness programs
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Recommendation 5: Expand Employee Training

Provide:

- Skill development programs
 - Technical certifications
 - Career advancement workshops
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13. Project Limitations

The analysis has several limitations:

- Correlation does not imply causation.
- External economic factors are not considered.
- Employee feedback surveys are not included.
- Some behavioral factors cannot be measured directly.

Future projects may incorporate predictive analytics and machine learning techniques for more advanced workforce forecasting.

14. Final Conclusion

The Adecco HR Analytics project successfully demonstrates how Excel-based analytics can be leveraged to understand employee behavior, workforce dynamics, and attrition patterns.

The analysis revealed that employee engagement, compensation, training opportunities, career growth, and work-life balance are among the most important factors affecting retention and workforce performance.

The organization has a relatively stable workforce, with an average employee tenure of approximately seven years. However, specific employee groups—including those with low job involvement, longer commuting distances, fewer benefits, and limited career growth opportunities—show higher attrition tendencies.

Overall Conclusion

Employee retention is not driven by a single factor. It is the result of compensation, engagement, career development, work-life balance, and organizational culture working together. By focusing on these areas, Adecco India can reduce attrition, improve employee satisfaction, strengthen workforce productivity, and achieve long-term organizational success.